Collector Task Manager

Tags	ctaskman, deadlock, align_orphan_devices, alignClusterCollector
Team Owner	The Collectables
Status	ACTIVE
Last Audit	Dec 4, 2020

- Introduction
- Diagnosis and Mitigation
  - proc_mgr is not restarting CTM
- Post Mortem Articles
  - 2023-08-16 EM-60307 Collector Task Manager Stops Responding After 5 Minutes
  - 2023-06-06 EM-58930 Deadlock between ctaskman and cisco VOS PP RBA code, config push trigger bottleneck
  - 2019-02-07 EPlus Unable To Onboard New Devices Outage (EM-29130)

Introduction

Diagnosis and Mitigation

proc_mgr is not restarting CTM

See this article: [2020-05-02 Symmetry: RBA proc fails II]

This was fixed in 10.2.0 (EM-34767). Its rather insidious. It boils down to the fact that the command "systemctl restart X" can hang indefinitely, and em7_patch_manager used to sometimes hang, causing the "systemctl restart X" command to hang, which is issued by proc_mgr for some of the processes it restarts. So proc_mgr then hangs (because it waits for "systemctl restart X" to finish, instead of running that through a subshell with a timeout), and

therefore stops restarting CTM. So after CTM exits (which occurs every hour), it is never restarted by proc_mgr.

The temporary mitigation is to kill -9 patch_manager. It might be another process though, so you need to verify which one is hung. Please read the full article linked above.

Post Mortem Articles

2023-08-16 EM-60307 Collector Task Manager Stops Responding After 5 Minutes

Link: EM-60307: Collector Task Manager Stops Responding After 5 Minutes CLOSED

In this case, the last debug message before CTM became unresponsive was "Pushing async tasks to remote appliances". The customer noticed this problem disappeared as a result of the mitigations in this case: Phone Home Collector Troubleshooting Guide | 2023 08 21 EM 6033 0: 30 out of 100 collector can suddenly no longer connect to ... . This would imply that CTM will hang when one or more of the phone home collectors is failing to connect to the CDB. If so, this should be regarded as a defect.

2023-06-06 EM-58930 Deadlock between ctaskman and cisco VOS PP RBA code, config push trigger bottleneck

Link: TTEC - Digitial ctaskmanager UHE deadlocks

In this case, there was a deadlock between an update to master_dev.legend_device by ctaskman (as called from the function align_orphan_devices) and by some cisco VOS PP RBA code (in the function alignClusterCollector). The design of the config push triggers makes this much more likely, and should be considered a bottleneck we will eventually need to fix.

@Michael Burkhard describes a work around in this comment: SOL-24635: Unhandled Exception - c_task_manager.py - SL1 WxCCE - cug_realign TO DO quoted below:

```
My recommendation is yes, leave it disabled.  That RBA is intended to make sure the VOS components and physical devices are on the same collector before they are merged.  Unless the customer is
discovering new VOS clusters, that RBA should not be needed.

If the customer is discovering new VOS clusters, I still don't think it is necessary.  From my local testing, it appears that the merge can be done successfully even when the two devices are on
separate collectors.  We put the collector realignment RBA in place to workaround issues last seen on the 8.x platform.  It may no longer be needed at all on the newer platform.

As a last resort, if the customer discovers a new VOS cluster, and the devices do not merge automatically, the customer can still merge them manually through the UI.

```

```
If this resolves the issue for the customer, this ticket should be converted to a defect so
we can update the VOS PP as needed.
```

2019-02-07 EPlus Unable To Onboard New Devices Outage (EM-29130)

Article here: [EPlus Unable To Onboard New Devices Outage]

In this case, a bad collector was identified that was causing CTM to hang when probing it. This brought the rest of CTM to a halt, preventing async jobs from being pushed to collectors.

The exact mechanism is not clear but Raynard proposed that a firewall was silently dropping the SYN packets from the CDB.

The CTM logs by themselves were not enough to identify that CU causing the problem. A drop file was created to add additional info. The logs were then inspected to find the bad CU, and that CU was moved out of any CUG so that CTM would stop responding to it.